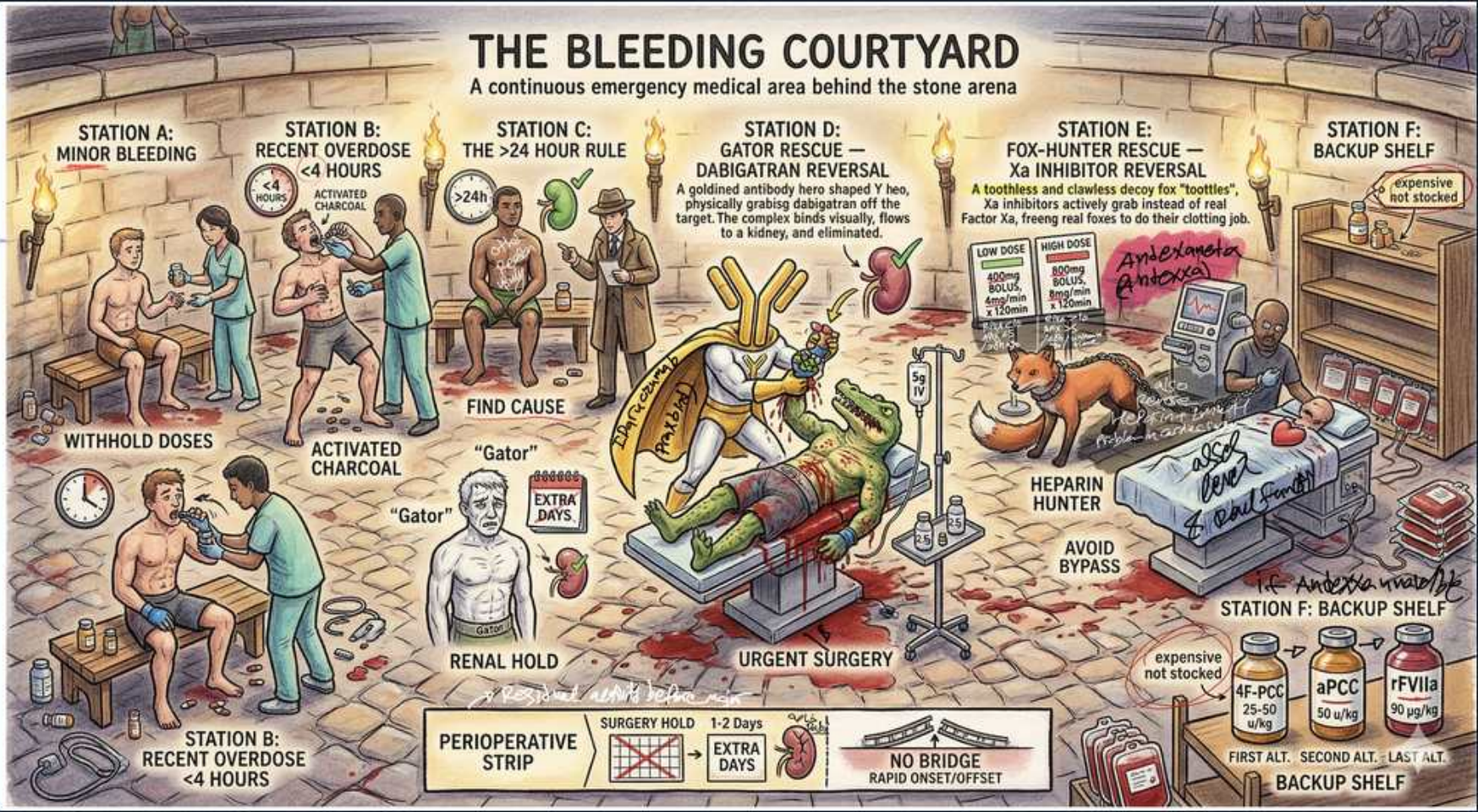


DOAC — The Bleeding Courtyard (Reversal & Bleeding Management)



1. Minor bleeding — withhold doses

WHAT YOU SEE

Station A patient on a bench, alarm clocks, 'WITHHOLD DOSES'

WHAT IT ENCODES

For MINOR/non-life-threatening bleeding on a DOAC the first move is simply to HOLD the next dose(s) and apply local hemostatic measures. DOACs have short half-lives (~8–17 h), so anticoagulant effect falls quickly once dosing stops — usually no reversal agent is needed. Supportive care, mechanical compression and treating the bleeding source are sufficient.

BOARD TRAP

Don't reach for andexanet/idarucizumab or PCC for trivial or mucosal bleeding — reversal agents are reserved for LIFE-THREATENING or critical-site bleeding. Over-reversing a stable patient adds thrombotic risk for no benefit.

2. Recent overdose <4 h — activated charcoal

WHAT YOU SEE

Station B '<4 HOURS', patient drinking 'ACTIVATED CHARCOAL'

WHAT IT ENCODES

If a DOAC was INGESTED within the last 2–4 hours (overdose or accidental excess), ORAL ACTIVATED CHARCOAL can reduce further GI absorption of the drug. This is an early, time-limited intervention aimed at lowering peak drug levels before they are fully absorbed.

BOARD TRAP

Charcoal only helps for RECENT ingestion (~<2–4 h). Past that window the drug is already absorbed and charcoal is useless — at that point management depends on bleeding severity and reversal agents, not decontamination.

3. >24-hour rule — find the cause

WHAT YOU SEE

Station C '>24h' hourglass, 'FIND CAUSE'

WHAT IT ENCODES

Once it is >24 hours since the last dose, most of a normally-cleared DOAC is gone, so persistent bleeding or a markedly abnormal coagulation test should prompt you to FIND ANOTHER CAUSE (renal failure causing drug accumulation, liver disease, a structural bleeding lesion, or concurrent antiplatelet/NSAID use) rather than blaming residual drug.

BOARD TRAP

After 24 h in a patient with normal renal function, don't assume ongoing 'DOAC effect' — look for accumulation (low CrCl) or an anatomic/second cause. Renal impairment is the classic reason a DOAC effect persists far longer than expected.

4. Dabigatran → idarucizumab

WHAT YOU SEE

Station D 'DABIGATRAN REVERSAL', glowing Y-shaped antibody fragment grabbing target, kidney

WHAT IT ENCODES

Dabigatran is a direct THROMBIN (factor IIa) inhibitor. Its specific antidote is IDARUCIZUMAB, a humanized monoclonal antibody FRAGMENT (Fab) that binds dabigatran with very high affinity and neutralizes it almost immediately (within minutes). Use it for life-threatening bleeding or before urgent/emergency surgery. Dabigatran is heavily RENALLY cleared, so accumulates in renal failure (and is partly dialyzable).

BOARD TRAP

Idarucizumab reverses ONLY dabigatran — it does NOT work on the -xabans (apixaban/rivaroxaban). Don't give andexanet for dabigatran; match the antidote to the mechanism (IIa inhibitor → idarucizumab, Xa inhibitor → andexanet).

5. Gator = dabigatran clue

WHAT YOU SEE

Green 'GATOR' face and gator figures, 'GATOR'

WHAT IT ENCODES

The recurring GATOR is the visual hook for dabigatran (the direct thrombin inhibitor). It anchors the rule that this is the ONE oral DOAC with an antibody antidote (idarucizumab) and the one most dependent on the kidney for clearance.

BOARD TRAP

Keep the gator (dabigatran, anti-IIa) separate from the -xaban factor-Xa inhibitors — they have different antidotes (idarucizumab vs andexanet) and different monitoring assays.

6. Renal hold / extra days

WHAT YOU SEE

Kidney 'RENAL HOLD', notepad 'EXTRA DAYS', perioperative strip

WHAT IT ENCODES

Because dabigatran is ~80% renally cleared, patients with reduced CrCl need the drug HELD for EXTRA DAYS before surgery (its half-life lengthens markedly as renal function falls). The perioperative strip at the bottom shows holding 1–2 days for normal renal function and longer for impaired clearance or high-bleeding-risk procedures.

BOARD TRAP

A 'standard' 24–48 h hold is NOT enough in renal impairment for dabigatran — failure to extend the hold based on CrCl is a classic perioperative bleeding error.

7. Urgent surgery on the table

WHAT YOU SEE

Patient (multicolored) on operating table, blood, 'URGENT SURGERY'

WHAT IT ENCODES

For EMERGENCY/urgent surgery in an anticoagulated patient who cannot wait for natural drug clearance, give the specific reversal agent first: idarucizumab for dabigatran, andexanet alfa (or 4-factor PCC if andexanet unavailable) for a -xaban. The goal is rapid restoration of hemostasis so the operation can proceed safely.

BOARD TRAP

Don't simply transfuse FFP hoping to 'reverse' a DOAC — FFP does not meaningfully reverse direct oral anticoagulants. Use the targeted antidote or PCC.

8. Xa inhibitor reversal — andexanet alfa

WHAT YOU SEE

Station E 'FOX-HUNTER RESCUE — Xa INHIBITOR REVERSAL', fox/ferret, decoy text

WHAT IT ENCODES

Apixaban and rivaroxaban (the -xabans) are direct FACTOR Xa inhibitors. Their specific antidote is ANDEXANET ALFA — a recombinant, catalytically INACTIVE decoy factor Xa that acts as a high-affinity bait: it binds and sequesters the Xa inhibitor (and TFPI), freeing real factor Xa to resume clotting. Used for life-threatening Xa-inhibitor bleeding (e.g., intracranial hemorrhage).

BOARD TRAP

Andexanet is a DECOY (inactive Xa), not active Xa — and it reverses only Xa inhibitors, not dabigatran. It is expensive, often non-stocked, and carries a thrombotic-risk signal, so 4F-PCC is a common alternative.

9. Heparin hunter / avoid bypass

WHAT YOU SEE

'HEPARIN HUNTER', 'AVOID BYPASS', handwritten notes

WHAT IT ENCODES

Caution note: andexanet alfa also binds and partly neutralizes HEPARIN, so after andexanet, heparin may not work — a major problem if the patient then needs cardiopulmonary BYPASS (which depends on heparin anticoagulation). Plan reversal strategy with the surgical/anesthesia team if bypass is anticipated; consider PCC instead in that setting.

BOARD TRAP

Giving andexanet to a patient who will imminently need heparin for bypass can leave you unable to anticoagulate them for the pump — a real, tested pitfall.

10. 4F-PCC alternative

WHAT YOU SEE

Backup shelf '4F-PCC 25–50', 'FIRST ALT'

WHAT IT ENCODES

FOUR-FACTOR PROTHROMBIN COMPLEX CONCENTRATE (4F-PCC, ~25–50 units/kg) contains factors II, VII, IX, X and is the FIRST-LINE ALTERNATIVE for Xa-inhibitor-associated life-threatening bleeding when andexanet is unavailable, too costly, or contraindicated. It bypasses the inhibited factor by flooding the system with clotting factors.

BOARD TRAP

PCC does not specifically 'neutralize' the DOAC — it overwhelms the block by supplying downstream factors. It is also the go-to when andexanet would interfere with subsequently needed heparin.

11. aPCC (FEIBA) — second alt

WHAT YOU SEE

Shelf 'aPCC 50 u/kg', 'LAST ALT'

WHAT IT ENCODES

ACTIVATED PCC (aPCC / FEIBA, ~50 units/kg) is a further fallback for DOAC-associated bleeding when 4F-PCC and andexanet are not options. It contains activated factors and bypasses the inhibited step, but carries a higher thrombotic risk, so it sits lower on the algorithm.

BOARD TRAP

aPCC is a later-line choice, not first-line — its higher prothrombotic potential means it is reserved for refractory cases.

12. rFVIIa — last resort

WHAT YOU SEE

Shelf 'rFVIIa', 'LAST ALT'

WHAT IT ENCODES

Recombinant activated factor VIIa is a last-resort hemostatic agent occasionally considered in catastrophic, refractory bleeding. Evidence for DOAC reversal is weak and thrombotic risk is significant, so it is essentially the bottom of the shelf.

BOARD TRAP

rFVIIa is NOT a validated DOAC antidote — do not pick it over idarucizumab/andexanet/4F-PCC for a DOAC bleed.

13. Perioperative strip — no bridging

WHAT YOU SEE

Bottom banner 'PERIOPERATIVE STRIP', 'NO BRIDGE — RAPID ONSET/OFFSET', 1–2 days / extra days

WHAT IT ENCODES

DOACs do NOT require heparin BRIDGING around procedures because of their rapid ONSET and OFFSET — you simply hold them for a defined interval (~1–2 days normal renal function, longer for high bleeding risk or impaired clearance) and restart afterward once hemostasis is secure. This is a key advantage over warfarin.

BOARD TRAP

Bridging a DOAC with LMWH is wrong and increases bleeding — bridging logic belongs to WARFARIN (slow offset/onset), not to short-acting DOACs.

14. Backup shelf empty / costly

WHAT YOU SEE

Station F 'BACKUP SHELF', empty shelves, 'EXPENSIVE NOT STOCKED'

WHAT IT ENCODES

Practical reality: specific DOAC antidotes (idarucizumab, andexanet) are EXPENSIVE and frequently NOT STOCKED in many hospitals. This is why supportive care, holding the drug, charcoal (if recent), and PCC remain central to real-world bleeding management — the 'shelf' may be bare when you need the branded agent.

BOARD TRAP

Don't plan management assuming the antidote is on hand — know the PCC-based fallback algorithm because andexanet/idarucizumab may be unavailable.